



TEMASEK JUNIOR COLLEGE
PRELIMINARY EXAMINATION
JC2 2021

CANDIDATE NAME

CIVICS GROUP

C	G			/	2	0
---	---	--	--	---	---	---

H1 BIOLOGY

Paper 2 Structured and Free-response Questions

8876/02

25 August 2021

2 hours

Candidates answer on the Question Paper

No Additional Materials are required.

READ THESE INSTRUCTIONS FIRST

Write your name and civics group in the spaces at the top of this page.

Write in dark blue or black pen.

You may use an HB pencil for any diagrams or graphs.

Do not use staples, paper clips, glue or correction fluid.

Answer **all** questions in the spaces provided on the Question Paper.

The use of an approved scientific calculator is expected, where appropriate.

You may lose marks if you do not show your working or if you do not use appropriate units.

The number of marks is given in brackets [] at the end of each question or part question.

For Examiner's use	
Section A	
1	/ 17
2	/ 18
3	/ 10
Section B	/ 15
Total	/ 60

Section A

Answer **all** the questions in this section.

- 1 Fig. 1.1 shows the synthesis of polypeptides on the rough endoplasmic reticulum.

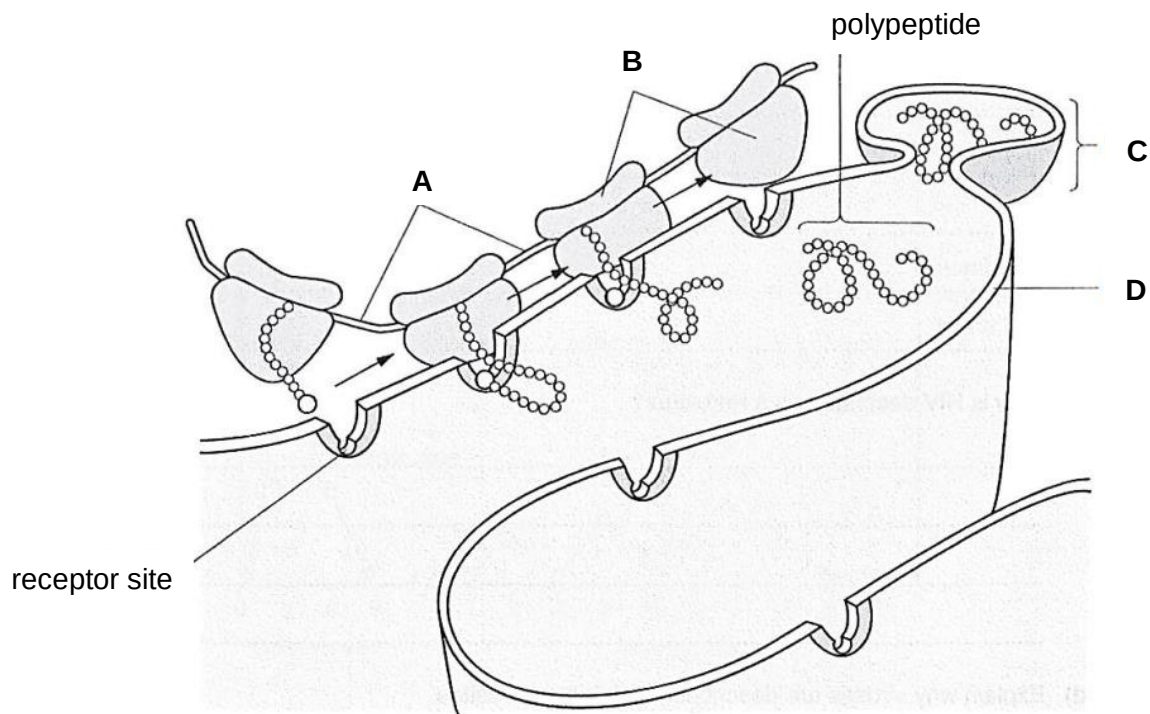


Fig. 1.1

- (a) Identify structures A, B, C and D, as shown on Fig. 1.1.

A: messenger RNA
 B: Ribosome / Ribosomal subunits
 C: (Transport) Vesicle
 D: Cisterna(e)

[4]

- (b) Use Fig. 1.1 to suggest how the structures labelled B attach to the endoplasmic reticulum through the growing polypeptide. [2]

- Free ribosome synthesises a polypeptide chain with an ER signal peptide.
- ER signal peptide attaches / binds to receptor site on endoplasmic reticulum bringing the ribosome closer to the endoplasmic reticulum, as shape of ER signal peptide is complementary to shape of receptor site.

An example of a polypeptide chain synthesised into the lumen of the endoplasmic reticulum is that of the hexosaminidase polypeptide chain. This will eventually give rise to the hexosaminidase enzyme, which is an enzyme found in lysosomes (lysosomal enzyme) that digests sphingolipids, a type of lipid.

(c) Describe what happens to the hexosaminidase polypeptide chain when it is released from the rough endoplasmic reticulum to eventually give rise to the hexosaminidase enzyme found in lysosomes. [3]

1. The hexosaminidase polypeptide chain folds into a globular 3D shape/conformation.
2. The ER vesicle containing the hexosaminidase protein buds off and travels along microtubules of the cytoskeleton.
3. ER vesicle then fuses with the cis-face of the Golgi apparatus (GA).
4. Hexosaminidase protein is chemically modified at the GA, forming hexosaminidase enzyme.
5. Hexosaminidase enzyme is packaged into vesicles and buds off from the trans-face of the GA, giving rise to lysosomes.

(d) Tay-Sachs disease is a genetic disorder that results in the destruction of nerve cells in the brain and spinal cord due to an accumulation of sphingolipids. It is normally caused by a genetic mutation in the hexosaminidase gene, which codes for the hexosaminidase enzyme.

One common mutation found on the hexosaminidase gene is the G444R mutation, where a glycine amino acid is replaced by an arginine amino acid at the 444th codon.

(i) State the type of mutation that has occurred. [1]

- Substitution mutation

The G444R mutation occurs at a splice site at the end of exon 11.

(ii) Explain how the G444R mutation can result in Tay-Sachs disease. [4]

1. Small nuclear RNAs (snRNAs) / small nuclear ribonucleoprotein (snRNPs) / spliceosome would be unable to recognise the splice site at the end of exon 11.
2. Hence, exon 11 will be removed / excised during RNA splicing.
3. The mRNA sequence is now changed, changing the amino acid sequence / primary structure of the polypeptide chain, such that the 444th codon is now arginine rather than glycine.
4. This in turn affects the folding of the polypeptide chain / enzyme, changing its 3D conformation / altering its secondary and tertiary structure
5. such that shape of active site is no longer complementary to shape of substrate/sphingolipid
OR
precise conformation/shape of the active site is lost.
6. Enzyme is no longer functional / unable to digest sphingolipids, leading to an accumulation of sphingolipids, resulting in Tay-Sachs disease.

OR

1. Glycine has a different R group from arginine.
2. Different R groups has different physical/chemical properties.
3. The changed amino acid occupies an important position in the amino acid sequence of the polypeptide.
4. This in turn affects the folding of the polypeptide chain / enzyme, changing its 3D conformation / altering its secondary and tertiary structure
5. such that shape of active site is no longer complementary to shape of substrate/sphingolipid

OR

precise conformation/shape of the active site is lost.

6. Enzyme is no longer functional / unable to digest sphingolipids, leading to an accumulation of sphingolipids, resulting in Tay-Sachs disease.

A proposed treatment for Tay-Sachs disease is enzyme replacement therapy, where a solution containing the functional hexosaminidase enzyme is injected into the bloodstream of the patient. However, this treatment method is not very effective, as the enzyme is unable to be taken into the lysosome.

(iii) Suggest why the hexosaminidase enzyme is unable to be taken into the lysosome. [1]

1. Enzyme is too large/polar to cross the hydrophobic core of the lysosomal membrane.

Tay-Sachs disease is a recessive condition that is commonly found in the Ashkenazi Jews population. Studies have found that this population is subjected to a stronger selection pressure by *Mycobacterium tuberculosis* (the bacterium that causes tuberculosis) due to crowded environments, and carriers of the mutated hexosaminidase allele have increased resistance towards *Mycobacterium tuberculosis*.

(e) With reference to the information provided, explain how the mutated hexosaminidase allele is preserved in the population. [2]

1. The mutated hexosaminidase allele is preserved via heterozygote advantage.
2. Heterozygotes / carriers are selected for as they do not die from Tay-Sachs disease or tuberculosis.
3. Upon reproduction, they are able to pass down the mutated hexosaminidase allele to their offspring.

[Total: 17]

- 2 Modern horses are a group of mammals adapted to living in the grassland. Their evolution can be reconstructed from the fossil record.

Table 2.1 shows the evolution of the teeth and height of the modern horse over 50 million years.

Table 2.1

Name	Teeth	Height / cm	Environmental conditions		
<i>Equus</i> (2 mya – today)	“Herbivorous teeth”	150	Predator present.	Grassland region. Mostly fibrous grass.	Edible insects extinct.
<i>Meryohippus</i> (26 mya)	Long cubical molars and flat grinding surfaces	100	Predator present.	Evolution of grass. Forest started to recede.	Edible insects becoming extinct.
<i>Mesohippus</i> (35 mya)	“Omnivorous teeth”	60	Predator present.	Forest filled with trees and shrubs (above 60cm).	Edible insects found on the ground and upper part of the shrubs.
<i>Eohippus</i> (50 mya)	Short, bumpy molars	25	Predator present.	Forest filled with trees and shrubs (above 60cm).	Edible insects found on the ground.

mya - million years ago

- (a) With reference to Table 2.1, explain how natural selection led to the evolution of the horse over time. [5]

1. There were selection pressures in the environment.
2. Genetic variation was present in the ancestor population due to random mutations.
3. Predator present and type of food were shrubs and insects.
4. Early horse ancestors were mostly small / short (25 or 60 cm) and had omnivorous teeth.
5. Such individuals were able to hide from predators and able to eat insects on the ground or upper part of the shrubs.
6. They were selected for.
7. However, environmental conditions changed.
8. Receding forest, evolution of grass and extinction of edible insects.
9. Individuals which were taller (e.g. 100 cm or 150 cm) and had herbivorous teeth
10. were able to escape from predator (e.g. run faster) and derive nutrients from grass.
11. They were selected for and so survived and reproduced to produce fertile, viable offspring.
12. Favorable alleles were passed to offspring resulting in higher frequencies of the alleles in the descendent population over time. These led to the modern horses which are tall and have herbivorous teeth.

With increasing global temperature due to rising carbon dioxide concentration in the atmosphere, one scientist hypothesises that grasses will benefit more than forest vegetation as they are better adapted to the temperature increase. A possible reason could be that the photosynthetic enzymes of grasses had an optimum temperature that is higher than the current average mid-day temperature of 30°C in the tropical grassland regions.

To test his hypothesis, the scientist carried out a study to study the effect of temperature on the growth of the grasses. Different patches of grass were subjected to different temperature inside experimental chambers. For all the experiments, other variables such as the species of grass, light intensity and nutrient solution were standardised.

At the end of the experiment, the changes in mass of the grass were calculated.

Table 2.2

Treatment	Mass of grass at the start of experiment / mg	Mass of grass at the end of experiment / mg	Change in mass as a percentage of the mass of grass at the start of experiment / %
30°C (Control)	1.24×10^4	1.53×10^4	23.4 or 23.39
36°C	1.43×10^4	1.87×10^4	30.8 or 30.77
42°C	1.44×10^4	1.74×10^4	20.8 or 20.83

(b) (i) Calculate the percentage change in the mass of grass and complete the last column of Table 2.2. [1]

(ii) In the space provided below, show all your working for the percentage change in the mass of grass at 42°C.

$$\text{Percentage change in the mass of grass at } 42^\circ\text{C} = \frac{1.74 \times 10^4 - 1.44 \times 10^4}{1.44 \times 10^4} \times 100\%$$

$$= 20.8\%$$

[1]

(iii) State the optimum temperature of the enzymes. [1]

- **36°C**

(iv) Discuss whether the results obtained by the scientist supported his hypothesis. [4]

1. The results supported his hypothesis at 36°C as the percentage increase in the mass of grass was greater than the control.
2. Increase of 30.8% (for 36°C) was greater than the increase of 23.4% (for control).
3. Increasing temperature increased the ATP production needed for growth as the kinetic energy of the photosynthetic enzymes and their substrates increase.
4. This increases the frequency of effective collision and hence the formation of enzyme-substrate complexes per unit time.
5. However, the result does not support his hypothesis at 42°C as the percentage increase in the mass of grass was lower than the control.
6. This is due to the denaturation of the enzymes as temperature exceeds the optimum temperature.
7. **Ref to** disruption of bonds (e.g. hydrogen bonds) that maintain the specific conformation of the active sites.

Any 1

8. Shape of active site no longer / less complementary to the substrate.
9. Lower frequency of enzyme-substrate complex formation.

The cell wall of plant cells is made up of mainly cellulose. Cellulose synthase is responsible for the synthesis of cellulose.

Fig. 2.3 shows the cross-section of the enzyme embedded within the cell surface membrane of plant cells.

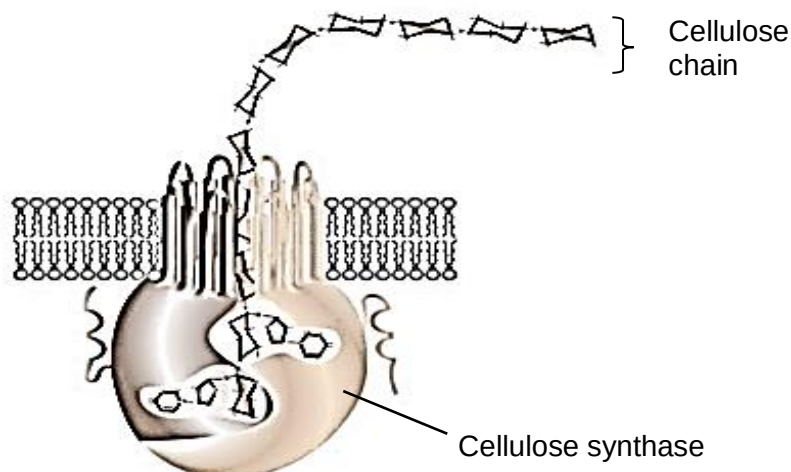
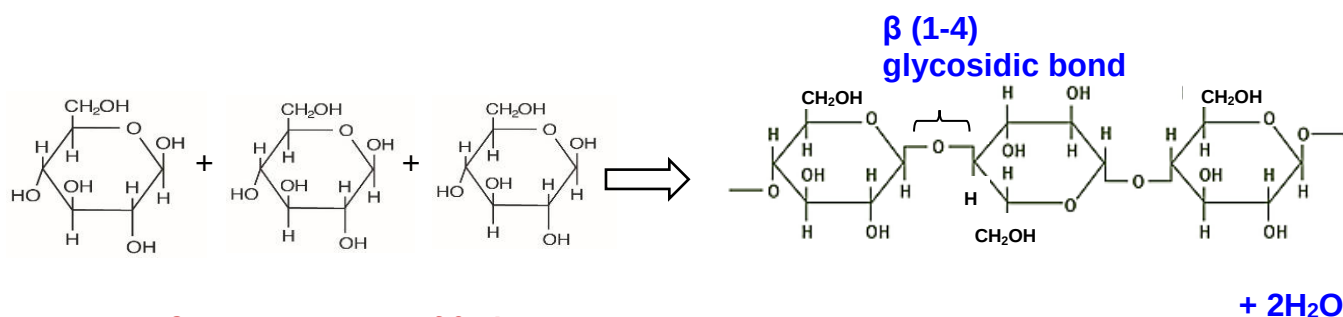


Fig. 2.3

(c) (i) In the space below, show how three monomers of cellulose are joined with each other to form a part of the cellulose chain.

Your answer should include the name of the bond that joins two adjacent monomers.



1. Correct structure of β glucose monomer
2. Correct structure of the 3 linked β glucose monomers
3. Alternate glucose rotated 180° to its adjacent monomer
4. 2 water molecules
5. Correct bond label

Cellulose synthase is part of a complex. Studies have shown that each complex is made up of six clusters, while each cluster consists of several cellulose synthase enzymes as shown in Fig. 2.4.

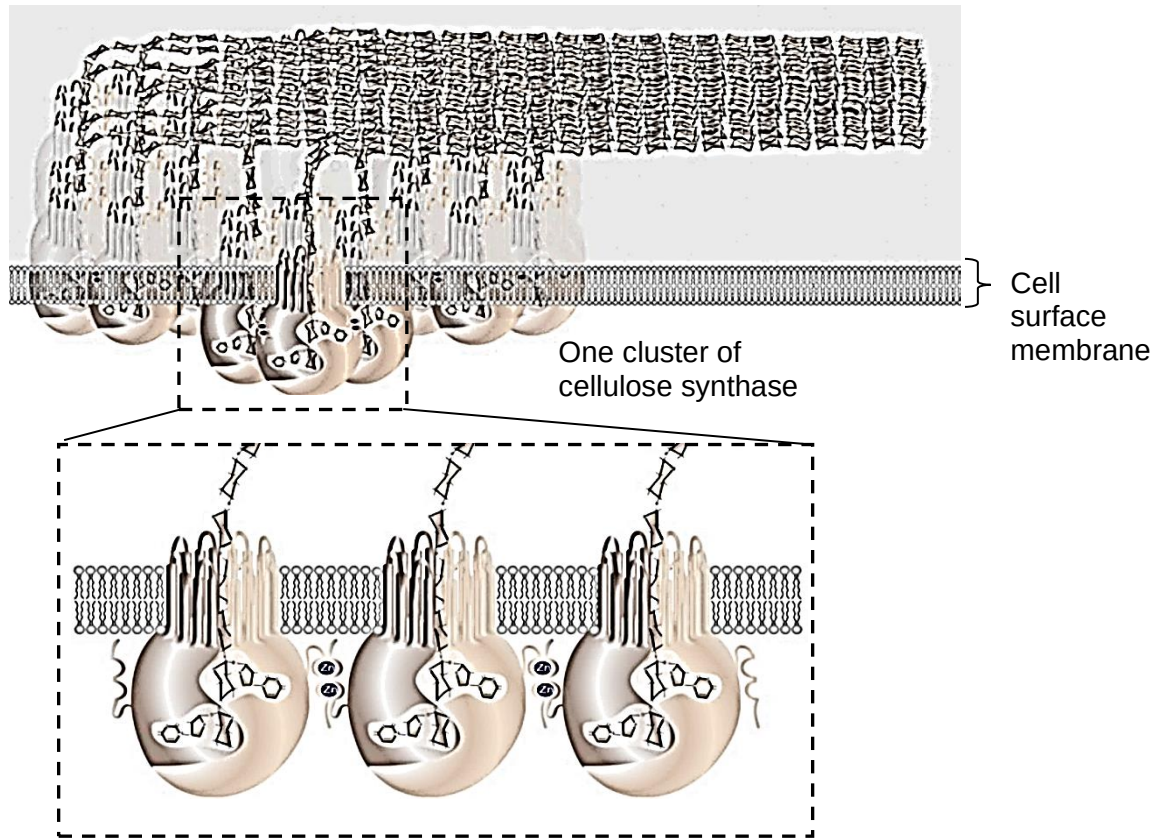


Fig. 2.4

(ii) With reference to Fig. 2.4 and the role of cellulose, explain why cellulose synthase enzymes tend to be clustered in close proximity with each other. [3]

1. Each cellulose synthase enzyme catalyses the synthesis of a straight chain of cellulose with the outward projection of hydroxyl groups.
2. Close proximity of the enzymes (in clusters) ensures that the straight cellulose chains to be aligned closely with each other so as to facilitate the formation of hydrogen cross-links required in the formation of microfibrils.
3. Microfibrils will then assemble to form fibers.
4. This will provide high tensile strength required of cellulose as a structural polysaccharide.

[Total: 18]

- 3 Antigens are found on human red blood cells and are used to determine the human ABO blood group system. The type and presence of antigens are determined by three alleles, I^A , I^B and I^O . Alleles I^A and I^B are co-dominant and both alleles are dominant to allele I^O .

(a) Define the term co-dominance using the example of the ABO blood group. [2]

1. Co-dominance occurs when both alleles are expressed equally in the phenotype of a heterozygote.
2. I^A and I^B alleles are both expressed equally as antigens A and B on the red blood cell membranes, giving rise to blood group AB.

There is another type of blood group system known as the Rhesus blood group, which is also genetically controlled. The gene for the Rhesus blood group has two alleles. The allele for Rhesus positive, Rh^+ , is dominant to that for Rhesus negative, Rh^- . Fig. 3.1 shows the inheritance of the Rhesus blood group in one family.

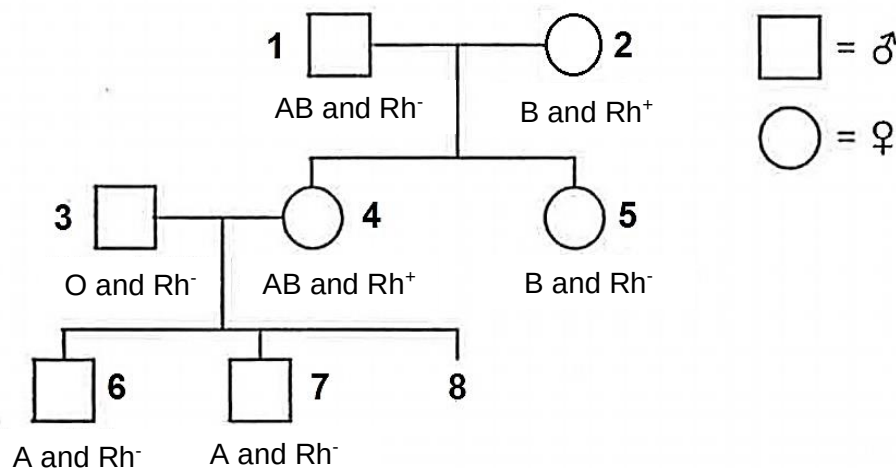


Fig. 3.1

- (b) (i) Explain one piece of evidence from Fig. 3.1 which shows that the gene for Rhesus blood group is **not** on the X chromosome. [2]

1. Individuals 6 and 7 would not be Rh negative.
2. This is because if this is sex-linked, they would have received the Rh-positive allele on the X chromosome from their mother (individual 4), making them Rh positive instead.

- (ii) Calculate the probability that individual 8 will be a boy with blood group B and Rh positive.

$$\begin{aligned} \text{Probability} &= \frac{1}{2} \times \frac{1}{2} \times \frac{1}{2} \text{ [1]} \\ &= \frac{1}{8} / 0.125 \text{ [1]} \end{aligned}$$

[2]

Fig. 3.2 shows the movement of chromatids in some blood cells that have just started metaphase of mitosis.

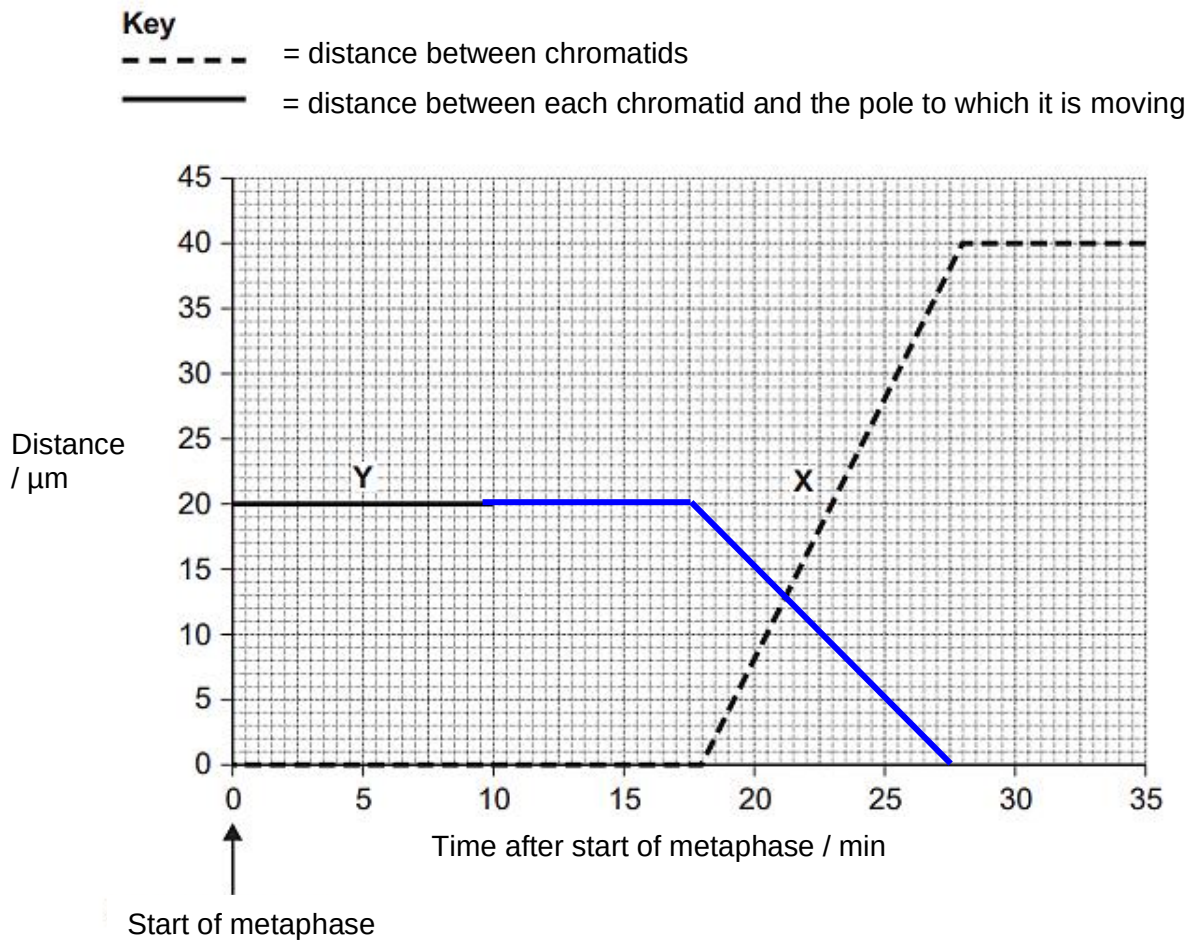


Fig. 3.2

(c) Complete line Y on Fig. 3.2 to show the expected distance between each chromatid and the pole to which it is moving. [2]

(d) Chromosomes also behave in a similar manner during metaphase I and II of meiosis. Explain how metaphase I and II of meiosis can result in genetic variation. [2]

1. There is independent arrangement of homologous chromosomes at the equator of the cell during metaphase I
2. and independent arrangement of chromosomes at the equator of the cell during metaphase II.
3. This leads to different combinations of chromosomes in daughter cells and new combination of alleles.

[Total: 10]

Section B

Answer **one** question in this section.

Write your answers on the lined paper provided at the end of this Question Paper.

Your answers should be illustrated by large, clearly labelled diagrams, where appropriate.

Your answers must be in continuous prose, where appropriate.

Your answers must be set out in parts **(a)** and **(b)** as indicated in the question.

4 **(a)** Outline the conversion of light energy to chemical energy by photosynthesis. [8]

1. Light energy is absorbed by photosynthetic pigments / chlorophyll during photophosphorylation / light reaction.
2. Light energy is used to energise and displace the electrons from special chlorophyll a / primary pigments in the photosystems.
3. Electrons are passed to the electron transport chain (ETC) / through ETC / series of electron carriers
4. until they are finally accepted by NADP^+ (final electron acceptor) to form NADPH.
5. Electrons are also used to synthesise ATP.
6. During the transfer of electrons through the ETC, energy is released to generate H^+ gradient across thylakoid membrane.
7. H^+ is pumped / actively transported from chloroplast stroma to thylakoid space.
8. H^+ is passively transported from thylakoid space back to stroma by facilitated diffusion.
9. H^+ diffuse through stalked particles containing ATP synthase which helps to synthesise ATP.
10. (Chemical energy in) ATP and NADPH are used in the Calvin cycle / dark reaction to synthesise carbohydrates.
11. Ribulose biphosphate (RuBP) combine with carbon dioxide to produce glycerate phosphate, catalysed by rubisco.
12. ATP and NADPH are used to reduce glycerate phosphate to triose phosphate / glyceraldehyde-3-phosphate.
13. Some of the triose phosphate are used to build other carbohydrates (e.g. glucose, starch, etc.) that stores the chemical energy that is converted from light energy.
14. Some of the triose phosphates and ATP are used to regenerate RuBP so that Calvin cycle can continue to occur.

(b) Discuss the ethical implications of stem cell research and how induced pluripotent stem cells (iPSCs) overcome some of the issues.

[7]

1. Killing embryo is similar to killing a life / disrespect for the value of human life.
2. Stem cells injected into patients may be rejected by patient's immune system.
3. The implanted stem cells may become cancerous.
4. The stem cells could be contaminated by viruses, which could harm health.
5. Treatment could potentially be costly, only available to the rich.
6. For treatments which involve gene editing of stem cells, this may result in attempts to change human traits not related with disease.
7. Using iPSC does not involve killing of embryos.
8. If donor cell is from patient, tissues derived from iPSCs will have identical match, which avoids immune rejection by the patient.
9. If donor cell is from patient, tissues derived from iPSCs will have identical match, which saves time waiting for suitable donor / reduces treatment time.

[Total: 15]

5 (a) Describe the multistep development of cancer.

[8]

1. Development of cancer is a multistep process that occur over time.
2. Accumulation of gene mutations in a single cell that affect the control of the cell cycle / checkpoints of the cell cycle.
3. Example: dysregulation of G1 checkpoint
4. Results in the transition from G1 to S phase despite absence of growth factors, nutrients and appropriate cell size.
5. Hence, during cell division, gene mutations are passed to daughter cells.
6. Mutations could arise due to excessive exposure to factors such as [at least 1 named example] chemical carcinogens (e.g. tar in cigarette smoke), ionising radiation (e.g. X-ray).
7. Gene mutations could also be inherited from parents (i.e. occurred in parent and the mutated copy of the gene / mutant allele was passed to offspring.)
8. Cancer cells do not undergo apoptosis and undergo uncontrolled cell division
9. The development of cancer includes angiogenesis whereby the cancer cells stimulate the growth of blood vessels towards themselves.
10. The blood vessels bring nutrients to the growing (cancerous / malignant) tumour to sustain its growth.
11. During metastasis, some cancer cells detach from the original tumor and enter the blood vessels.
12. They are transported to other parts of the body where they develop into secondary tumours.

[Turn over]

(b) Explain the significance of membrane fluidity in biological processes.

[7]

Membrane fluidity refers to phospholipids and membrane proteins being always in constant motion / moving laterally in the plane of the membrane.

[Protein secretory pathway]

1. Membrane fluidity is significant to protein transport within the cell / out of the cell / attachment of proteins in the cell surface membrane.
2. **Ref to** membrane-bound vesicles forming and fusing with membrane of organelle or cell surface membrane.

[Any one]

3. Allows for (transport) vesicles forming / budding from rough endoplasmic reticulum (rER) to Golgi apparatus.

OR

Allows for (transport) vesicles forming / budding from the cis-face to trans-face of Golgi apparatus

OR

Allows for (secretory) vesicles forming / budding from trans-face of Golgi apparatus to cell surface membrane

4. Any named example

[Secretion out of cell] E.g. antibodies by plasma cells, insulin by beta cells via exocytosis

[Attachment of proteins in cell surface membrane] E.g. viral glycoproteins of influenza/HIV virus

[Transport across membrane by bulk transport]

5. Membrane fluidity is significant for endocytosis / phagocytosis / pinocytosis / receptor-mediated endocytosis,
6. Involves the invagination of cell surface membrane
7. Cell surface membrane to extend outwards / form pseudopodia to surround particles
8. Any named example -
Entry of influenza virus by human respiratory epithelial cells / Entry of pathogens by white blood cells / immune cells / AVP

[Transport across membrane by diffusion or active transport]

9. Membrane fluidity is significant for the formation of transient pores required for simple diffusion of small and non-polar molecules directly across the membrane.

OR

to allow for osmosis of water molecules directly across the membrane.

10. Allows for transport / channel / carrier proteins to be embedded in the cell surface membrane and to undergo conformational change for facilitated diffusion / active transport of molecules across the membrane.
11. Any named example: active transport of H^+ ions by proton pumps / facilitated diffusion of H^+ ions by ATP synthase (only awarded if example is linked to correct process)

[Respiration / photosynthesis]

- 12. Membrane fluidity is significant for oxidative phosphorylation and photophosphorylation to take place
- 13. **Ref to** conformational change of electron carriers for electrons to be passed down electron carriers of decreasing energy levels

[Digestion of substances]

- 14. Membrane fluidity is significant for formation of lysosomes from trans-face of Golgi apparatus.
- 15. Membrane bound lysosome able to fuse with endocytic vesicles / endosomes for autophagy / phagocytosis / digestion/break down of substances.

[Cytokinesis]

- 16. Membrane fluidity is significant for cytokinesis.
- 17. **[Plant cells]** Golgi vesicles to fuse to form the cell plate in plant cells
- 18. **[Animal cells]** Allows for invagination / pinching in of the cell surface membrane in animal cells, forming the cleavage furrow

[Total: 15]